

CM-3 Exploratory Data Analysis

3TLDE.A13 - Législation Numérique et Enjeux de l'Analyse des Données



May 26, 2026

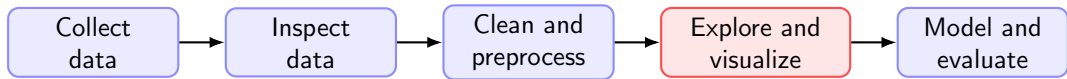
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Where are we in the data analysis workflow?



In CM-1, we learned how to load, inspect, and summarize a dataset.

In CM-2, we learned how to make the data usable: missing values, duplicates, outliers, types, encoding, and scaling.

Today

After the data is usable, we need to understand it. Exploratory Data Analysis helps us discover distributions, relationships, and useful patterns before modeling.

Cleaning and EDA are related, but not the same

CM-2: Cleaning

- Are values missing?
- Are there duplicated rows?
- Are data types correct?
- Are categories consistent?
- Are some values impossible?

CM-3: EDA

- What does each variable look like?
- Which variables are related?
- Which variables are related to the target?
- Are there groups with different behavior?
- What hypotheses can we form?

Important distinction

Cleaning asks: **Can we use the data?**

EDA asks: **What does the data tell us?**

What is Exploratory Data Analysis?

Exploratory Data Analysis (EDA) is the process of **understanding** a dataset using:

- descriptive statistics;
- tables and grouped summaries;
- visualizations;
- simple comparisons between variables.

The goal is not only to produce plots.

Core idea

EDA means asking questions about the data, choosing appropriate summaries or visualizations, and interpreting the results carefully.

Typical questions in EDA

Before training a model, we often ask:

- What is the distribution of each numerical variable?
- Are some variables highly skewed?
- Are there extreme values that may influence the analysis?
- Which categories are frequent or rare?
- Does the target variable depend on some features?
- Are some features strongly related to each other?
- Are there interactions between several variables?

Practical message

EDA is useful because it helps us make better decisions for preprocessing, feature engineering, and modeling.

A running example: the Titanic dataset

We will use the Titanic dataset as a running example.

Each row represents one passenger.

Column	Meaning	Type
Survived	survival indicator: 0 or 1	categorical / target
Pclass	ticket class: 1, 2, or 3	categorical
Sex	passenger sex	categorical
Age	passenger age	numerical
Fare	ticket fare	numerical
Embarked	port of embarkation	categorical

The main target is:

Can we understand which factors are associated with survival?

Learning objectives of CM-3

At the end of this lecture, you should be able to:

- explain the role of EDA in the data analysis workflow;
- identify numerical and categorical variables;
- choose appropriate plots according to variable types;
- perform univariate analysis for numerical and categorical variables;
- study relationships between a feature and a target variable;
- use grouped analysis and correlation to explore multivariate patterns;
- write clear interpretations from tables and visualizations;
- connect EDA findings to preprocessing and modeling decisions.

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A simple EDA workflow

A practical EDA workflow can be organized in four levels.

1. Dataset overview: rows, columns, variable types, target



2. Univariate analysis: one variable at a time



3. Bivariate analysis: two variables, often feature vs target



4. Multivariate analysis: several variables together

Remember

EDA usually goes from simple to more complex: first understand variables individually, then study relationships.

Starting an EDA in pandas

A common first step is to inspect the structure of the dataset.

```
1 import pandas as pd
2
3 # Load the dataset
4 df = pd.read_csv("titanic_cleaned.csv")
5
6 # Basic structure
7 print(df.shape)
8 print(df.columns)
9 print(df.info())
10
11 # First rows
12 print(df.head())
```

This gives us:

- the number of rows and columns;
- column names;
- data types;
- a first view of the values.

Numerical summaries with describe()

For numerical variables, `describe()` gives a compact summary.

Statistic	Age	Fare
count	891	891
mean	29.7	32.2
std	13.0	49.7
min	0.4	0.0
25%	22.0	7.9
50%	29.0	14.5
75%	35.0	31.0
max	80.0	512.3

Interpretation

The maximum fare is much larger than the median fare. This suggests a strongly **skewed distribution** or **extreme values**.

Categorical summaries with `value_counts()`

For categorical variables, we usually count categories.

```
1 df["Sex"].value_counts()  
2 df["Sex"].value_counts(normalize=True) * 100
```

Sex	Count	Percentage
male	577	64.8%
female	314	35.2%

Interpretation

There are more male passengers than female passengers in the dataset.

Choosing an appropriate plot

Situation	Useful plots	Example question
One numerical variable	histogram, KDE, boxplot	How is Age distributed?
One categorical variable	bar chart, countplot	How many passengers per class?
Numerical vs categorical	boxplot, violin plot	Does Fare differ by class?
Categorical vs categorical	grouped bar, stacked bar	Does survival differ by sex?
Numerical vs numerical	scatter plot, correlation	Is Fare related to Age?

Common mistake

A plot can be technically correct but analytically useless if it does not answer a clear question.

EDA is not only visualization

Visualizations are important, but EDA also includes:

- summary statistics;
- grouped tables;
- percentages and rates;
- comparison of distributions;
- interpretation in natural language.

A good EDA result should often have this form:

The survival rate is higher for female passengers than for male passengers. This suggests that Sex is likely to be an important predictive feature for survival.

A weak EDA statement would be: *I plotted survival by sex.*

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Univariate analysis: one variable at a time

Univariate analysis studies one variable independently of the others.

For a numerical variable, we usually want to know:

- Where are the values concentrated?
- Is the distribution symmetric or skewed?
- Are there extreme values?
- What are the typical values?

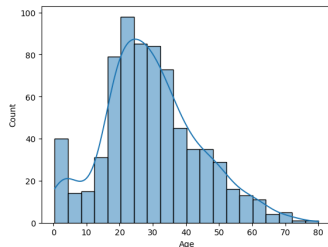
For a categorical variable, we usually want to know:

- Which categories exist?
- Which categories are frequent or rare?
- Are some categories dominant?

Histogram: understanding a numerical distribution

A histogram divides numerical values into intervals and counts how many observations fall in each interval.

```
1 import seaborn as sns
2 import matplotlib.pyplot as plt
3
4 sns.histplot(data=df, x="Age", bins=20, kde=True)
5 plt.title("Distribution of Age")
6 plt.show()
```



What to look for

Look at the global shape: concentration, spread, skewness, multiple peaks, and possible unusual values.

Interpretation

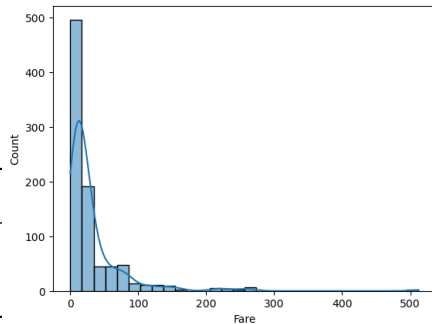
The dataset mostly contains young and middle-aged adults (between 20 and 40 years old). Fewer children and elderly passengers.

Mean and median: why both are useful

Consider two variables:

- Age: often moderately spread;
- Fare: often strongly right-skewed.

Variable	Mean	Median	Interpretation
Age	29.7	29.0	mean and median are close
Fare	32.2	14.5	mean is much larger than median



Interpretation

When the mean is much larger than the median, a few very large values may pull the mean upward.

Skewness: an intuitive view

A distribution is **skewed** when values are not balanced around the center.

Symmetric distribution

- left and right sides are similar;
- mean and median are close;
- example: many measurement errors around a central value.

Right-skewed distribution

- many small or medium values;
- a few very large values;
- mean is often larger than median;
- example: income, ticket fares, house prices.

EDA consequence

For skewed variables, median and quartiles are often more informative than the mean alone.

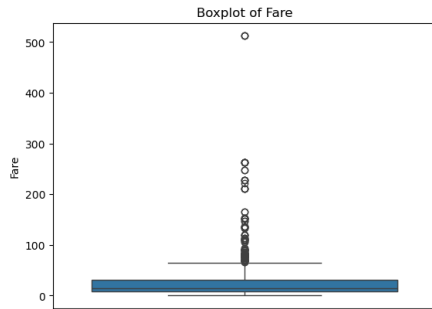
Boxplot: summary and potential outliers

A boxplot summarizes a numerical variable using quartiles.

```
1 sns.boxplot(data=df, y="Fare")  
2 plt.title("Boxplot of Fare")  
3 plt.show()
```

It helps us see:

- the median;
- the central part of the distribution;
- the spread of the values;
- potential outliers.



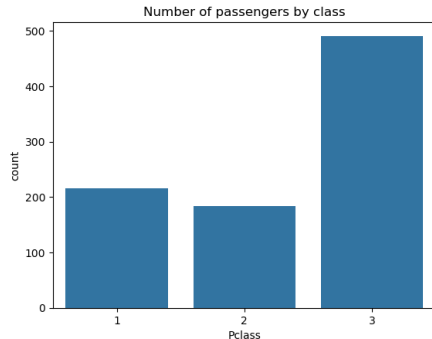
Connection with CM-2

In CM-2, boxplots were used to detect suspicious values. In CM-3, we also use them to understand the shape and variability of a variable.

Countplot: understanding a categorical variable

For a categorical variable, the basic question is:
how many observations are in each category?

```
1 sns.countplot(data=df, x="Pclass")
2 plt.title("Number of passengers by class")
3 plt.show()
```



Interpretation example

If third-class passengers are much more numerous than first-class passengers, **raw survival counts may be misleading**. We should compare survival rates by class.

Rare categories

Some categories may contain very few observations.

This matters because:

- percentages for small groups can be unstable;
- a model may not learn reliable patterns for rare categories;
- rare categories may need to be grouped, depending on the task.

Example

If a port of embarkation has only 2 passengers, a survival rate of 100% does not necessarily mean that this port is truly safer. The group is too small for a stable conclusion.

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Bivariate analysis: two variables together

Bivariate analysis studies the relationship between two variables.

In supervised learning, we often focus on:

feature $\longrightarrow$ **target**

For Titanic:

- Does Sex seem related to Survived?
- Does Pclass seem related to Survived?
- Does Age differ between survivors and non-survivors?
- Does Fare differ between survivors and non-survivors?

Categorical feature vs target: survival rate by sex

We can compute the mean of a binary target.

```
1 df.groupby("Sex")["Survived"].mean()
```

If Survived is coded as 0 or 1, then: mean of Survived in a group = survival rate of the group.

Sex	Survival rate
female	0.74
male	0.19

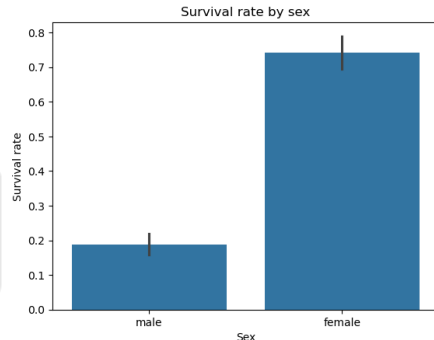
Interpretation

Female passengers have a much higher survival rate than male passengers in this dataset. This suggests that Sex is strongly associated with survival.

Visualizing survival rate by group

A bar plot can show the average value of a binary target by group.

```
1 sns.barplot(data=df, x="Sex", y="Survived")
2 plt.title("Survival rate by sex")
3 plt.ylabel("Survival rate")
4 plt.show()
```



Interpretation

The height of each bar is the average of Survived in that group, so it represents the survival rate.

Do not write only: *the bars are different*. Write what the difference means.

Counts are not rates

A common mistake is to compare only raw counts.

Sex	Total passengers	Survivors	Survival rate
female	314	233	74.2%
male	577	109	18.9%

Important

The number of male survivors is not enough to understand survival. Since there are many more male passengers, we must compare survival rates.

Categorical feature vs target: survival by class

```
1 df.groupby("Pclass")["Survived"].mean()
```

Passenger class	Survival rate
1	0.63
2	0.47
3	0.24

Interpretation

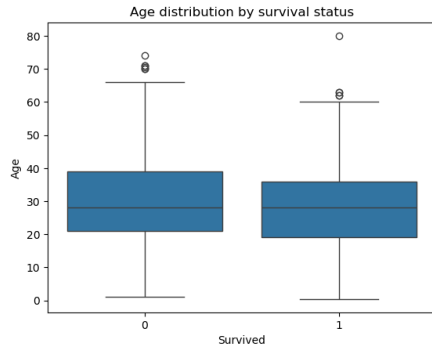
First-class passengers have a higher survival rate than third-class passengers. The passenger class may capture social or spatial differences on the ship.

Careful: EDA shows association, not direct causality.

Numerical feature vs target: Age and survival

To compare a numerical variable across target groups, we can use a boxplot.

```
1 sns.boxplot(data=df, x="Survived", y="Age")
2 plt.title("Age distribution by survival status")
3 plt.xlabel("Survived")
4 plt.ylabel("Age")
5 plt.show()
```



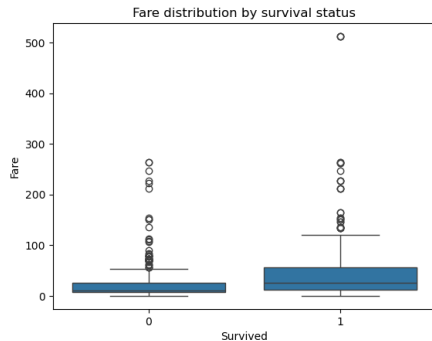
Questions to ask

Are the medians different? Is one group more spread out? Are there many young or old passengers in one group?

Numerical feature vs target: Fare and survival

Ticket fare is often strongly skewed, so a boxplot can reveal group differences and extreme values.

```
1 sns.boxplot(data=df, x="Survived", y="Fare")
2 plt.title("Fare distribution by survival status")
3 plt.xlabel("Survived")
4 plt.ylabel("Fare")
5 plt.show()
```



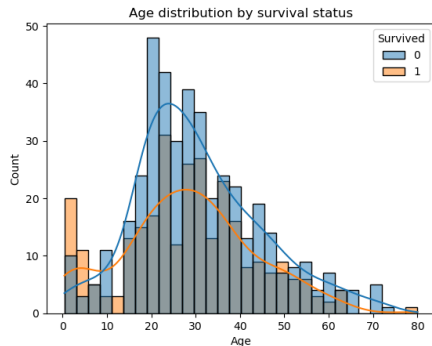
Possible interpretation

If survivors have higher fares on average, fare may be associated with survival. However, fare is also related to passenger class, so the relationship may not be direct.

Using histograms with hue

Histograms can compare distributions between groups.

```
1 sns.histplot(  
2     data=df,  
3     x="Age",  
4     hue="Survived",  
5     bins=30,  
6     kde=True  
7 )  
8 plt.title("Age distribution by survival status")  
9 plt.show()
```



When useful

This is useful when we want to see whether two groups occupy different ranges of a numerical variable.

EDA reveals associations, not causality

EDA can show that two variables are related in the dataset.

But it usually cannot prove that one variable causes the other.

Example

Survivors may have higher fares on average. This does not necessarily mean that paying more directly caused survival. Fare is related to passenger class, cabin location, and other factors.

A more careful statement is:

*Fare is associated with survival in this dataset.
This association may be partly explained by passenger class.*

Good and bad interpretations

Weak interpretation	Better interpretation
The plot is interesting.	The survival rate is much higher for female passengers than for male passengers.
There are outliers.	A few passengers paid very high fares, which makes the fare distribution strongly right-skewed.
Class matters.	Survival rate decreases from first class to third class, suggesting that passenger class is associated with survival.
Fare causes survival.	Fare is associated with survival, but this may be confounded with passenger class.

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Why multivariate analysis?

A relationship observed with two variables may change when we consider a third variable.

Example questions:

- Is Sex still related to survival within each passenger class?
- Is the relation between Fare and survival actually due to Pclass?
- Do younger passengers have different survival rates depending on class and sex?

Main idea

Multivariate EDA helps us detect interactions and possible confounding effects.

Grouped analysis with two categorical variables

We can group by two variables before computing the survival rate.

```
1 df.groupby(["Sex", "Pclass"])["Survived"].mean()
```

Sex	Pclass	Survival rate
female	1	0.97
female	2	0.92
female	3	0.50
male	1	0.37
male	2	0.16
male	3	0.14

The grouped table suggests two patterns:

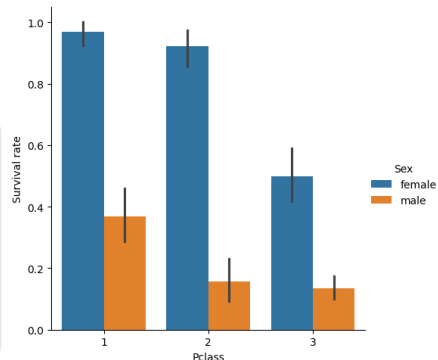
- within each class, female passengers have higher survival rates than male passengers;
- within each sex, first-class passengers often have higher survival rates than third-class passengers.

Interpretation

Visualizing grouped analysis

A grouped bar plot can represent the survival rate by class and sex.

```
1 sns.catplot(  
2     data=df,  
3     x="Pclass",  
4     y="Survived",  
5     hue="Sex",  
6     kind="bar"  
7 )  
8 plt.ylabel("Survival rate")  
9 plt.show()
```



How to read it

Compare bars within the same class to study the effect of sex. Compare bars of the same color across classes to study the effect of class.

Pivot tables for compact summaries

A pivot table is often easier to read than a long grouped table.

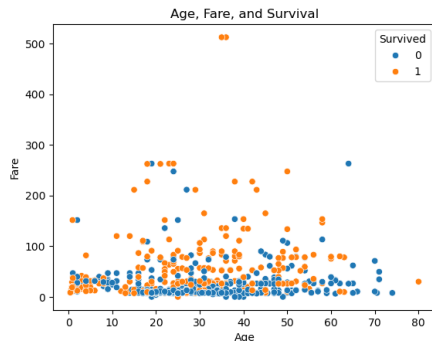
```
1 pivot = df.pivot_table(  
2     values="Survived",  
3     index="Pclass",  
4     columns="Sex",  
5     aggfunc="mean"  
6 )  
7 print(pivot)
```

Pclass	female	male
1	0.97	0.37
2	0.92	0.16
3	0.50	0.14

Scatter plot: numerical vs numerical

A scatter plot represents each observation as a point.

```
1 sns.scatterplot(data=df, x="Age", y="Fare", hue="Survived")
2 plt.title("Age, Fare, and Survival")
3 plt.show()
```



Questions to ask

Is there a visible trend? Are there clusters? Are extreme fares concentrated in one survival group?

Correlation: the main idea

Correlation measures the strength of a linear relationship between two numerical variables.

The Pearson correlation coefficient is usually between -1 and 1 :

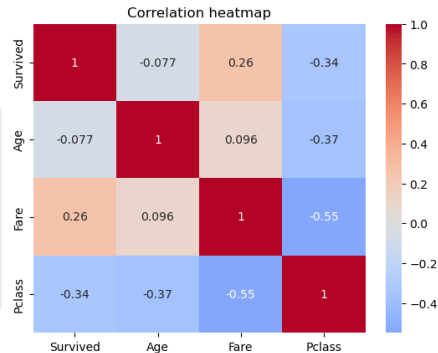
- close to 1 : strong positive linear relationship;
- close to -1 : strong negative linear relationship;
- close to 0 : weak linear relationship.

Important limitation

Correlation mainly captures linear relationships between numerical variables. A low correlation does not always mean that there is no useful relationship.

Correlation matrix and heatmap

```
1 # num_df = df.select_dtypes(include="number")
2 # corr = num_df.corr()
3
4 corr = df[['Survived', 'Age', 'Fare', 'Pclass']].corr()
5
6 sns.heatmap(corr, annot=True, cmap="coolwarm", center=0)
7 plt.title("Correlation heatmap")
8 plt.show()
```



How to use it

A correlation heatmap gives a quick overview of linear relationships among numerical variables. It can help identify redundant variables or variables related to the target.

Correlation: common mistakes

- Correlation does not imply causation.
- Correlation is sensitive to outliers.
- Correlation may miss non-linear relationships.
- Correlation between encoded categorical variables may be hard to interpret.

Example

If `Pclass` is coded as 1, 2, 3, a correlation with survival can be computed, but the result depends on the numerical coding of a categorical concept.

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EDA informs the next steps

EDA is useful because it suggests concrete actions.

EDA observation	Possible next step
Fare is strongly skewed	consider log transformation
Sex is strongly related to survival	keep it as an important feature
Pclass and Fare are related	check redundancy or interaction
Some categories are rare	group rare categories if needed
Outliers exist in Fare	investigate, cap, transform, or keep
Categorical features exist	encode them before modeling

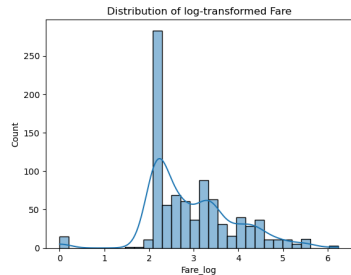
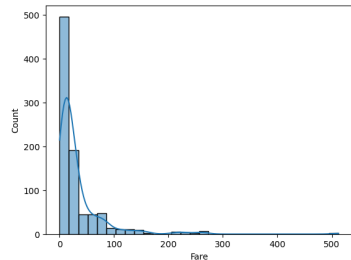
Example: log transformation for skewed variables

If a numerical variable is strongly right-skewed, a log transformation may make it easier to model.

```
1 import numpy as np
2
3 df["Fare_log"] = np.log1p(df["Fare"])
4
5 sns.histplot(data=df, x="Fare_log", bins=30)
6 plt.title("Distribution of log-transformed Fare")
7 plt.show()
```

Why log1p?

$\log1p(x)$ computes $\log(1 + x)$. It is useful when some values can be zero.



EDA and feature engineering

Feature engineering means creating or transforming variables to make useful information more explicit.

Examples from Titanic:

- create `FamilySize` = `SibSp` + `Parch` + 1;
- create `IsAlone` from `FamilySize`;
- extract title from passenger name: Mr, Mrs, Miss, etc.;
- transform `Fare` using logarithm;
- group rare categories if they are too small.

Practical message

EDA often gives ideas for better features. But each new feature should have a clear reason.

Example: creating FamilySize

Possible question:

Did passengers travelling alone have different survival rates from passengers travelling with family?

```
1 df["FamilySize"] = df["SibSp"] + df["Parch"] + 1
2
3 df.groupby("FamilySize")["Survived"].mean()
```

Family Size	1	2	3	4	5	6	7	8	11
Survival rate	0.303	0.553	0.578	0.724	0.200	0.136	0.333	0	0

Interpretation

This feature is not directly given in the raw data. It is constructed from existing columns because it may represent a meaningful social situation.

EDA should be reproducible

EDA is exploratory, but it should still be organized.

A good notebook should contain:

- clear section titles;
- code cells that can be re-run;
- plots with readable titles and axes;
- short interpretations after important outputs;
- a final summary of main findings.

Notebook principle

A reader should understand not only what you computed, but also why you computed it and what you concluded.

A simple EDA report structure

A short EDA report can follow this structure:

- 1 Dataset overview
- 2 Variable types
- 3 Univariate analysis
- 4 Relationship with the target variable
- 5 Multivariate patterns
- 6 Main findings
- 7 Possible next preprocessing or modeling steps

Final objective

The goal is to move from raw observations to clear conclusions that can guide the rest of the project.

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Key takeaways

- ① EDA is the bridge between data cleaning and modeling.
- ② The choice of visualization depends on the variable types and the analytical question.
- ③ Univariate analysis helps understand individual variables.
- ④ Bivariate analysis helps study relationships between variables and the target.
- ⑤ Multivariate analysis helps reveal interactions and possible confounding effects.
- ⑥ EDA suggests useful preprocessing, feature engineering, and modeling decisions.
- ⑦ A good EDA must include interpretation, not only code and plots.